



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

OOP: Object-Oriented Programming.



Learning Outcome:

Describe common activities at work to a using short sentences in familiar spoken and written situations.

Make the result report from the analysis of the information system, based on the customer requirements, following established rules and protocols.

Can Do Statements:

- I can explain briefly what Object Oriented Programming is (OOP).
- I can talk about the evolution of OOP.
- I can give examples of OOP.

Ask your partner:

- Do you think there is a universal language to program websites, web-based apps and programs? Why?
- What do think the difference between Java and JavaScript is?





Let's Explore!

John and Laura are talking about programming languages.



Instructions: Read the text below.

- John:** I talked to the new programmer this morning, he told me about a conference he attended last month in Japan. He is planning to present the latest tendency in programming languages next week. He mentioned OOP.
- Laura:** What do you mean by that?
- John:** It means object-oriented programming, that is, a programming model that divides the program responsibilities into objects and classes. Depending on the programming language you use, the syntax and semantics can change. However, it focuses on the objects you want to operate rather than the logic the program requires to manipulate them. You can perceive this model as a logical procedure that takes input data, processes it, and produces output data. This model reduces development time and ensures more accuracy while coding.
- Laura:** I get it. I had to read about OOP's history for a presentation last week; it originated as the dominant programming methodology in the early 60's and became popular in the mid-1990s. I read that the Xerox PARC laboratory implemented object-oriented programming to create their Smalltalk software.
- John:** That's right! They used OOP to design Smalltalk as well as Microsoft implemented C++ to relate the classes and objects within their programs. The programmer also told me about some of the different languages that companies implemented long ago and the in-demand ones that are trend nowadays; perhaps you used them or heard of them. For instance, Python, C, C++, C#, Java, Ruby, Delphi, or PHP.
- Laura:** I heard many of them were really common back in the day, like C, but now the ones that people implement the most to create, for example, websites or apps are JavaScript, Java, and C#. For example, we had to use Java to create applications on android and JavaScript to make a dynamic website for the company last time.



Let's Practice!

Let's talk about the meaning of OOP.

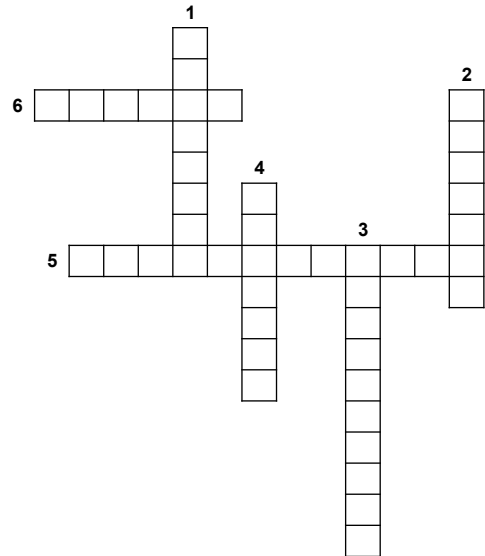
Instructions: Read the clues. Complete the crossword.

Down

1. OOP is a programming model that divides the program _____ into objects and classes.
2. It focuses on the _____ you want to operate.
3. rather than the logic required to _____ them.
4. This _____ model is a logical procedure that takes input data, processes it, and produces output data.

Across

5. This model reduces _____ time.
6. and ensures more accurate _____.



Let's Practice!

Laura is listing some examples of OOP.

Instructions: Select the correct answer.

1. Microsoft implemented C++ and C# to _____ windows apps.
 - a. write
 - b. create
 - c. send
 - d. teach
2. We had to use _____ some mobile applications on android as it's the core language it uses.
 - a. Linux to compile
 - b. JavaScript
 - c. Java to create
 - d. C++ to
3. Originally the used Java to _____ mobile apps before smartphones were out in the market.
 - a. developing
 - b. developed
 - c. develop
 - d. developer
4. Most developers used PHP before JavaScript to make webpages _____.
 - a. dynamic
 - b. dynamo
 - c. dynamism
 - d. dynamics
5. Software engineers _____ C to code their programs, now there are better versions of it, such as C++ and C#.
 - a. shared
 - b. presented
 - c. implemented
 - d. communicated
6. OOP languages _____ drastically within the first 3 decades since the 60's.
 - a. created
 - b. evolved
 - c. develop
 - d. grow





Extension Activity

Let's find out more about OOP.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about OOP languages.
- Describe their uses, similarities and differences.
- Write a short list with some examples.
- Show your list to your classmates and Instructor.

Self-Assessment

- | | Yes | No |
|---|-----------------------|-----------------------|
| 1. I look for information about OOP languages. | ☐ | ☐ |
| 2. I describe how they work. | ☐ | ☐ |
| 3. I write a short list comparing them. | ☐ | ☐ |
| 4. I share my list with my classmates and Instructor. | ☐ | ☐ |



Target Words

- **back in the day** (expression): longtime ago; in the past
e.g. I heard many of them were really common back in the day, like C, but now the ones that people implement the most to create, for example, websites or apps are JavaScript, Java, and C#.
- **coding** (noun): the action to write code
e.g. This model reduces development time and ensures more accurate coding.
- **attend** (verb): to go to an event, place
e.g. he told me about a conference he attended last month in Japan.
- **tendency** (noun): inclination or habit
e.g. He is planning to present the latest tendency in programming languages next week.
- **class** (noun): A class is a template name of an object in object oriented programming.
e.g. It means object-oriented programming, that is, a programming model that divides the program responsibilities into objects and classes.



Learning Points

Present Simple vs Past Simple Review.

You can use present simple and past simple to express ideas about permanent situations or habits that either happened at a specific time in the past or commonly happen in the present.

PRESENT SIMPLE

Use the **present simple** tense to express those ideas that are general or routinely in the present.

In order to make a sentence in the present simple form, use the following structures:

Affirmative Form: **Subject + Verb + Complement.**

Negative form: **Subject + Do / Does + Not + Verb + Complement.**

Interrogative form: **Do / Does + Subject + Verb + Complement + ?**

Remember that for the third person in affirmative form, you have to add S, ES, IES, at the end of the verb.

Examples:

- *I **make** use of OOP languages to create excellent websites and programs.*
- *Jane **doesn't know** how to work with C#.*
- ***Do** you **know** what OOP stands for?*

PAST SIMPLE

Use the past simple tense to talk about events that finished at a specific point in the past.

Remember that the structures you have to use to make sentences or question in this time are the following:

Affirmative Form: **Subject + Verb past form + Complement.**

Negative form: **Subject + Did + Not + Verb + Complement.**

Interrogative form: **Did + Subject + Verb + Complement + ?**

Examples:

- *They **popularized** the term OOP long time ago.*
- *I **didn't know** what OOP mean, but now I do.*
- ***Did** you **know** that OOP languages are more than common that what you think?*

Examples

- *It originated as the dominant programming methodology in the early 60's.*
- *I heard that the Xerox PARC laboratory implemented object-oriented programming to create their Smalltalk software.*





Learning Points

The 9 Most in Demand Programming Languages.

Taking into account that OOP languages are there since the early 60's, there are hundreds of programming languages that are object oriented. Here you will find a top 9 of the most used ones.

Below you will find a chart with the most in demand programming languages.

OOP LANGUAGE	DESCRIPTION
<i>SQL</i>	SQL (pronounced 'sequel') Programming language most apps, pc programs, and web-based apps use to store data or retrieve data from databases.
<i>JAVA</i>	JAVA is one of the most common programming languages programmers use to develop apps; as a matter of fact, all native android apps are JAVA-based. Nowadays. More than 9 million apps are based on JAVA, as it is easy to program and apps last long periods of time.
<i>JAVASCRIPT</i>	Programming language that differs from Java because its main purpose is to make webpages interactive. Nowadays, most browsers support JavaScript as a substantial part of the websites.
<i>C #</i>	(pronounced C-sharp) Most programs that run under .Net Framework, that is, windows based apps, need C# to run, as it is the most common language to write windows apps along with C++.



Learning Points

C++	(pronounced C-plus-plus) C, C++, and C# are the common languages windows developers and programmers use to write their programs, apps and games, as it works under the .Net Framework.
PYTHON	Python is another famous programming language that some institutions think easier to start with when learning to program, that is why many schools and institutions rather teach it first than Java.
PHP	PHP, along with JavaScript, is one of the most common scripting languages that programmers use to create dynamic webpages..
IOS / swift	Programming language for native apple apps (iOS and OX), reliable and similar to C++ codes.
RUBY	Ruby is an all-purpose programming language that is best used to program web-based apps.